

MPAIRWE LAUBEN

SOFTWARE ENGINEERING STUDENT · CLOUD-NATIVE · ML

ABOUT

Software engineering student at Makerere University (Expected graduation: Jan 2027, CGPA: 4.38 / 5.0). Proficient in Linux system administration, Python (Django/ Flask), Laravel, React, and Git, with practical hands-on experience in machine learning (NumPy, Pandas, scikit-learn), Docker, Kubernetes, and mobile development (Flutter, React Native). Active across cloud platforms, emergency dispatch backends, and applied ML systems.

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WORK EXPERIENCE

AIBOS Uganda
Kampala
Feb 2024 – Aug 2026

DEVSECOPS & SYSTEMS ENGINEER

- Administer Linux server environments (Ubuntu/ Debian), writing automated Bash scripts for system maintenance, cron automation, and service supervision
- Containerize microservices using Docker and orchestrate workloads on Kubernetes with Pod Security Standards, NetworkPolicies, and Helm charts
- Own CI/CD workflows on GitHub Actions with disciplined Git branching, cutting deploy times ~60% while integrating container vulnerability scans
- Run infrastructure-as-code (Terraform) across cloud environments with automated security policy gates on every deployment
- Deploy and maintain Prometheus, Grafana, and Loki monitoring stacks with SLO-driven alerts and incident runbooks
- Enforce compliance posture, encryption, RBAC, and centralized secret management across infrastructure

LINUX · BASH · DOCKER · KUBERNETES · GIT · GITHUB ACTIONS · TERRAFORM · PROMETHEUS · GRAFANA · AWS

Sauti Health
Kampala
2026 – Present

DEVELOPER · ENGINEERING PARTNER

- Partner with Sauti Health on backend systems powering a pre-hospital emergency dispatch platform for sub-Saharan Africa
- Build high-throughput backend services in Python (FastAPI/Django) and Linux environments coordinating responders over USSD on 2G networks
- Develop offline-first mobile workflows using Flutter and React Native with resilient sync and conflict resolution for low-bandwidth zones
- Package and deploy microservices with Docker containers for reproducible test and production environments
- Integrate mobile-money APIs (MTN MoMo, Airtel Money) and automated reporting pipelines for healthcare interoperability
- Maintain Git-based CI/CD pipelines and system telemetry to ensure continuous 24/7 dispatch uptime

PYTHON · DJANGO/FASTAPI · LINUX · DOCKER · FLUTTER · REACT NATIVE · GIT · POSTGRESQL · USSD/IVR

SKILLS

TRIVY GRYPE SNYK COSIGN SIGSTORE
SLSA OPA / GATEKEEPER KYVERNO
FALCO VAULT EXTERNAL SECRETS
POD SECURITY STANDARDS
NETWORKPOLICIES GITHUB ACTIONS
ARGOCD FLUX TERRAFORM PULUMI
MLFLOW WEIGHTS & BIASES DVC FEAST
KUBEFLOW RAY TRITON BENTOML
EVIDENTLY SCIKIT-LEARN TENSORFLOW
PYTORCH PANDAS NUMPY LANGCHAIN
LANGGRAPH MODEL CONTEXT PROTOCOL
VERCEL AI GATEWAY AI SDK V6 QDRANT
PGVECTOR BRAINTRUST PROMPTFOO
LANGSMITH DEEPEVAL GUARDRAILS
LINUX (UBUNTU/DEBIAN/RHEL)
BASH SCRIPTING DOCKER KUBERNETES
HELM AWS GCP AZURE VERCEL
PROMETHEUS GRAFANA LOKI
OPENTELEMETRY ISTIO POSTGRESQL
MYSQL MONGODB REDIS SNOWFLAKE
FIREBASE SUPABASE
PYTHON (DJANGO / FLASK) LARAVEL
REACT GIT NEXT.JS TYPESCRIPT PHP
LIVEWIRE TAILWIND FASTAPI
SHADCN/UI NODE.JS SPRING BOOT
FLUTTER REACT NATIVE KOTLIN SWIFT
APP STORE / PLAY

IONATECH
Kampala
Feb 2023 – Present

CO-FOUNDER · ML & SYSTEMS ENGINEER

- Co-founded IONATECH (ionatec.com) — developing applied ML, data analysis, and intelligent cloud systems for East African enterprises
- Build machine learning and data analysis pipelines with Python using NumPy, Pandas, and scikit-learn for tabular classification and regression
- Develop LLM and agentic workflows using LangChain, LangGraph, and Model Context Protocol (MCP) with rigorous evaluation benchmarks
- Containerize ML models and API endpoints with Docker for standardized deployment on Linux hosts
- Lead technical mentorship, conducting peer code reviews on Git and guiding junior developers in production practices

PYTHON · NUMPY · PANDAS · SCIKIT-LEARN · DOCKER · LINUX · GIT · LANGCHAIN · FASTAPI

Makerere University
Kampala
2022 – Expected Jan 2027

SOFTWARE ENGINEERING STUDENT

- Engineered Smart Discussion Forum — a collaborative academic platform built with Laravel, Livewire, and React components with Tailwind CSS
- Developed backend web APIs and data services in Python using Django and Flask, emphasizing ORM modeling and automated unit testing
- Administered Linux laboratory servers, creating Bash shell scripts for task automation, system diagnostics, and cron scheduling
- Implemented predictive models and statistical analysis using Python (NumPy, Pandas, scikit-learn) on academic datasets
- Built and deployed cross-platform mobile applications in Flutter and React Native with Firebase real-time synchronization
- Maintained version control across multiple student and faculty projects using collaborative Git workflows

PYTHON · DJANGO · FLASK · LARAVEL · REACT · GIT · LINUX · NUMPY · PANDAS · SCIKIT-LEARN · FLUTTER

LINKS

PORTFOLIO
mpairwelauben.me

LINKEDIN
mpairwe-lauben-3a1461344

GITHUB
github.com/mpairweLandwind

GITHUB
github.com/mpairwe7

IONATECH
ionatec.com

SAUTI HEALTH
sautihealth.com

LANGUAGES

English (Fluent)

German (Basic)

PAHAPPA LIMITED

Kampala

Jun 2022 – Jan 2024

DEVOPS INTERN → ENGINEER

- Built enterprise applications with Spring Boot (MVC, Data JPA, Security, Cloud)
- Used Hibernate ORM/JPA with MySQL, PostgreSQL, Oracle, and MongoDB databases
- Designed Microservice and Monolithic architectures based on project requirements
- Developed RESTful APIs and event-driven systems for high-throughput workflows

SPRING BOOT · JAVA · HIBERNATE · MYSQL · POSTGRESQL · MICROSERVICES · REST API

WEB INFO-NET LIMITED

Kampala

Jun 2018 – Aug 2022

APPLICATION DEVELOPER INTERN & TRAINER

- Developed mobile apps using Swift, Java, and Kotlin for iOS and Android platforms
- Built an Odoo-based Employment Management System for HR automation
- Implemented secure authentication and third-party API integrations
- Provided Level 1 technical support, troubleshooting, and end-user training

SWIFT · JAVA · KOTLIN · ODOO · IOS · ANDROID

ZALONIDENTALHUB

Kampala

Apr 2022 – Present

LEADING DEVELOPER & MAINTAINER

- Led full SDLC for dental practice management software from conception to delivery
- Built C#/VB.NET desktop systems for inventory and patient management
- Developed a Flutter mobile app with e-commerce, appointments, and booking features
- Ensured seamless desktop-mobile data interoperability
- Led a team of developers, defining software architectures and coding standards

C# · VB.NET · FLUTTER · DESKTOP · MOBILE · TEAM LEAD

PERSONAL DETAILS

DATE OF BIRTH
22 October 1995

NATIONALITY
Ugandan

LOCATION
Kampala, Uganda

EDUCATION

Makerere University

2022 – Expected Jan 2027
CGPA 4.38

Bachelor of Science in Software Engineering

- Software Engineering student at Makerere University (Expected Jan 2027) with a current CGPA of 4.38 / 5.0
- Specialised in Systems Architecture, Applied Machine Learning, and Enterprise Software Engineering
- Engineered full-stack applications in Laravel and React with role-based access control (RBAC)
- Developed backend web services, automation scripts, and REST APIs using Python (Django, Flask)
- Applied machine learning and data analysis pipelines using NumPy, Pandas, and scikit-learn
- Practical coursework and lab practice in Linux System Administration, Computer Networks, and Distributed Systems
- Certified academic transcript, UACE, and UCE pass slips verified and available

Makerere University

2019 – 2023

Bachelor of Science in Computer Engineering

- Transitioned to the Software Engineering programme
- Studied Digital Electronics and hardware design principles
- Hardware repair, maintenance, and systems diagnostics

Uganda Technical College – Lira

2016 – 2018
CGPA 4.77

National Diploma in Information Communication Technology

- Database Management and Web Development
- Internet technologies and networking fundamentals
- RBAC and Encryption implementation
- Java programming and object-oriented design
- Networking and Server Configuration

Boston High School – Mpala

2010 – 2015

UCE & UACE (O-Level & A-Level Certificates)

- Completed both Uganda Certificate of Education (UCE) and Uganda Advanced Certificate of Education (UACE)
- Graduated in Top 5% of class nationally
- Varsity Debate Team Captain

REFERENCES

Dr. Kimbugwe Nasser
Lecturer & Academic Supervisor
Makerere University, School of Computing & IT

Kalema Martin
Supervisor
AIBOS Uganda

Dr. Mark Magumba
Senior Lecturer
Makerere University, School of Computing & IT

Fahadi Nguyenguye
Technical Lead
PAHAPPA LIMITED

Dr. Ggaliwango Marvin
Lecturer & AI/ML Mentor
Makerere University, School of Computing & IT

CERTIFICATIONS

ONGOING	**AWS Certified DevOps Engineer · Professional**	— Amazon Web Services
ONGOING	**Certified Kubernetes Security Specialist**	— CNCF
ONGOING	**Certified Kubernetes Administrator**	— CNCF
ONGOING	**HashiCorp Certified · Terraform Associate**	— HashiCorp
PLANNED	**AWS Solutions Architect · Associate**	— Amazon Web Services
PLANNED	**Google Cloud Professional ML Engineer**	— Google Cloud